

Science · LAUNCH · Autumn 1 · Week 08 · SCI-LAUNCH-A1-W08-L3 · Date _____ Amylase Rate, Graph and Topic 1 Consolidation Today I can... ☐ Show relative-rate working using $1 \div \text{time}$ with the unit s^{-1} ☐ Read or plot the pH-rate pattern and limit the conclusion to the evidence ☐ Link a method limitation to an improvement, then distinguish digestion from diffusion Feedback: ☐ VF ☐ WS ☐ I ☐ NS ☐ E ☐ R _____	
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Amylase Rate, Graph and Topic 1 Consolidation · Route card

Reference only · use the task and sentence stem in your book.

Arrival · four short answers in your book

1. Explain why 90 s gives a larger relative rate than 180 s under matched conditions.
2. At 60 s starch is detected; at 90 s it is not. What endpoint value is used in the rate calculation?
3. Calculate relative rate for an endpoint of 120 s; show the equation, substitution and unit.
4. Distinguish DESCRIBE from EXPLAIN when answering about a graph.

Starter

Matched amounts, same iodine endpoint: reaction A reaches the endpoint in 90 s; B takes 180 s. Which is faster?

Exit ticket

Under the heading Exit ticket: answer the questions in your book.

1. Calculate both first-series relative rates and use them for a bounded comparison.
2. Use the repeat or sampling interval to give one limitation and a justified next investigation change.

Independent

1. Calculate relative rate for pH 6 and pH 7, showing equation, substitution and units. Check that shorter time gives a larger value. [4 marks]
2. Complete the five-point graph using the two calculated rates and three supplied points. Label both axes and state the rise-and-fall pattern. [4 marks]
3. Identify the highest tested activity in the first series using one numerical comparison. Explain how pH can affect the enzyme's active site. [3 marks]
4. Give one strength, one limitation and one linked improvement for the endpoint method. Use the separate pH 7 repeat of 120 s to qualify your confidence. [4 marks]
5. Distinguish enzyme digestion from diffusion in the supplied two-step diagram. Explain why movement from high to low concentration does not itself break starch into sugars. [3 marks]

Supported

1. Use $1 \div \text{time}$ to calculate the pH 6 rate from 120 s and the pH 7 rate from 90 s.
2. Add pH 6 and pH 7 to the supplied rate graph. Match each pH to its calculated rate; the other three points are supplied.
3. Choose the highest rate in the first series. Complete "In these tested conditions, pH ____ was fastest because ____."
4. The sampling interval is 30 s. Choose a limitation and a linked improvement: uncertain exact endpoint / use a shorter fixed interval if authorised.
5. The supplied diagram shows starch breaking into sugars, then particles moving from high to low concentration. Label the two changes: digestion / diffusion.

Choice

1. Match 120 s and 90 s to 0.00833 s^{-1} and 0.0111 s^{-1} . Show $1 \div \text{time}$ for each. Add the pH 6 and pH 7 points to the supplied graph; choose its pattern: rises then falls / always rises. Record C beside your book response.
2. Choose the highest first-series rate: pH 4 / pH 7 / pH 8. Give one number. Choose how pH may change activity: active-site fit / enzyme choice. Match the 30 s sampling gap to uncertain exact endpoint / exact timing; choose a shorter authorised interval / invented time as the improvement.

3. The pH 7 repeat took 120 s. Choose confidence: variation needs repeats / one exact optimum is proved. Match starch breakdown / net particle movement to digestion / diffusion. Write one sentence explaining the difference.

Aspire

1. Calculate pH 6 and pH 7 relative rates. Compare the first pH 7 rate with its separate 120 s repeat, retaining both results. [4 marks]
2. Complete the five-point first-series graph and place the separate repeat at the same pH 7 x-position with a distinct symbol. Explain why it is not pH 9 or a silently replaced point. [4 marks]
3. Evaluate “pH 7 is the exact optimum for every amylase”. Use the tested range, the repeat and the difference between data and a molecular explanation. [4 marks]
4. Give a strength–limitation–improvement chain. Explain which uncertainty a shorter authorised sampling interval reduces and which it leaves. [4 marks]
5. Explain why “diffusion digests starch” confuses two processes. Use the supplied definitions, and state why an enzyme-rate graph alone cannot show how sugars cross a gut membrane. [4 marks]

Staff mark I, S or C beside this task.

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LAUNCH SCIENCE | last lesson |

Enzyme rates and evidence

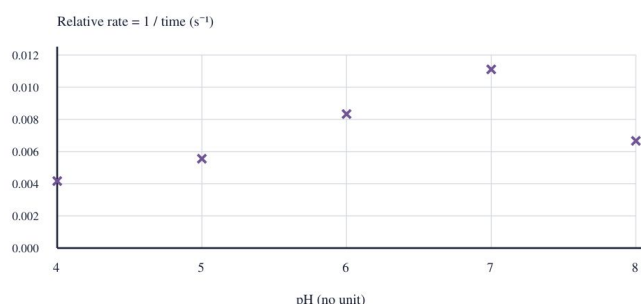
Knowledge organiser

Learning objective

I can calculate and graph relative amylase rates, judge what the evidence supports, and use enzyme and transport ideas in an exam-style explanation.

Success criteria

1. Show relative-rate working using $1 \div \text{time}$ with the unit s^{-1} .
2. Read or plot the pH–rate pattern and limit the conclusion to the evidence.
3. Link a method limitation to an improvement, then distinguish digestion from diffusion.

**Calculate a reciprocal**

Relative rate = $1/\text{time}$. For 120 s: $1/120 = 0.00833 \text{ s}^{-1}$. For 90 s: $1/90 = 0.0111 \text{ s}^{-1}$. A shorter endpoint time gives a larger reciprocal, for matched amounts and the same stopping rule.

Describe then qualify

These illustrative first-series points rise from pH 4 to 7 and fall at pH 8. pH 7 is highest among the values tested here. A separate pH 7 repeat gives 120 s (0.00833 s^{-1}); it shows variation, not a new pH value.

Explain and evaluate

pH can alter substrate binding and catalysis. The graph does not directly show active-site shape or prove permanent denaturation. Matched conditions support comparison; 30 s sampling limits precision. A shorter feasible authorised interval narrows that timing window.

Transfer the science

Enzyme digestion chemically breaks large molecules into smaller ones. Diffusion is net particle movement from higher to lower concentration. Diffusion does not itself chemically break starch into sugars.

Made by Matt | Use the reference to explain, calculate and evaluate.

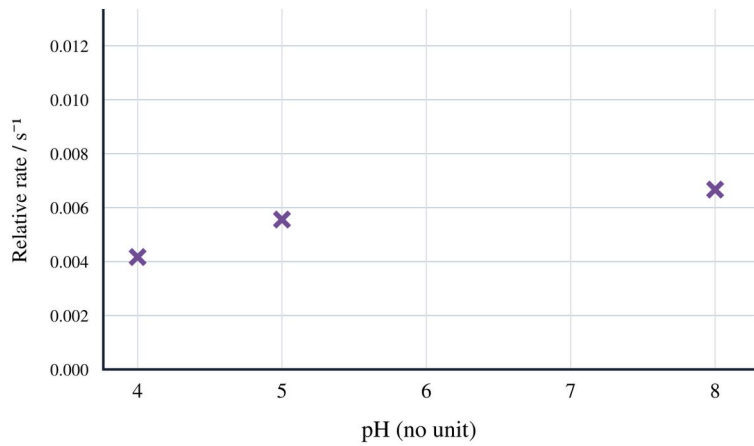
Practical reference data and models

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Enzyme: biological catalyst; speeds a reaction without being used up. Substrate: substance it acts on. pH: a measure of acidity or alkalinity. Amylase acts on starch, producing smaller sugars.

pH	Endpoint / s	Relative rate / s^{-1}
4	240	0.00417
5	180	0.00556
6	120	0.00833
7	90	0.0111
8	150	0.00667

Supplied illustrative observations — not your practical results. Single runs; equal starting amounts and the same iodine endpoint. Samples every 30 s. pH 7: blue-black at 30/60 s, orange-brown at 90 s. Record 90 s.



Digestion: chemical breakdown



Starch chain → smaller sugars

Diffusion: net movement



Higher → lower concentration

Amylase catalyses starch digestion. Moving particles does not itself split their molecules.

Complete the reference graph: pH 4, 5 and 8 are shown; add pH 6 and pH 7. Aspire: add the separate repeat with a distinct symbol.

Rate: $1 \div \text{endpoint time}$. At pH 7: $1 \div 90 = 0.0111 \text{ s}^{-1}$ (3 significant figures). These are single-run rates, not means. Highest among tested: pH 7; exact optimum is not established. The extra pH 7 repeat (120 s) is a separate trial, not silently averaged.

Topic 1 transfer: digestion breaks large molecules into smaller molecules. Diffusion is net particle movement from higher to lower concentration; it does not break a starch molecule apart. Enzymes are catalysts, not living organisms. pH changes can alter activity; slower timing alone does not prove permanent denaturation.

Describe = give a pattern; explain = link cause and effect; evaluate = use evidence and limits. An unobserved endpoint by 300 s establishes no exact reciprocal rate.

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Staff: cut these apart before the lesson.

Move these pieces for the task. Record the answer and reason in your book.

Reciprocal value: Endpoint 120 s gives $1/120 = 0.00833 \text{ s}^{-1}$.	Direction check: The shortest time should correspond to the largest reciprocal rate.
Highest tested rate: The largest first-series relative rate occurs at pH 7 among pH 4–8 tested here.	Rise then fall: First-series relative rate rises from pH 4 to pH 7, then falls at pH 8.
Subjective endpoint: Different observers may judge a borderline iodine colour differently.	Shorter interval: A shorter fixed sampling interval could narrow the endpoint-time window if the authorised method allows it.
Preserve anomaly: One repeat is far from the others; keep it visible, check the method and justify any treatment.	